

Biochemistry lab

Experiment (10)

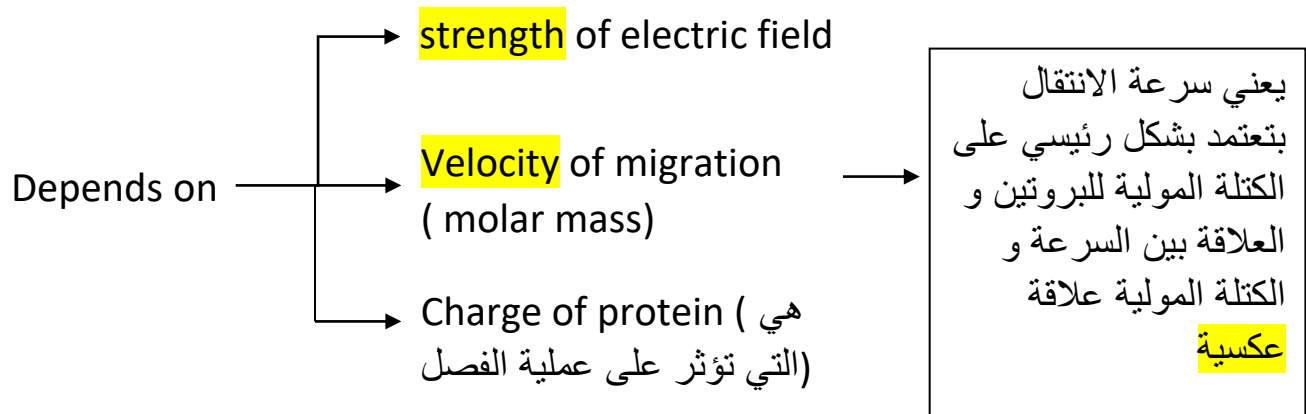


Electrophoresis

separation of proteins

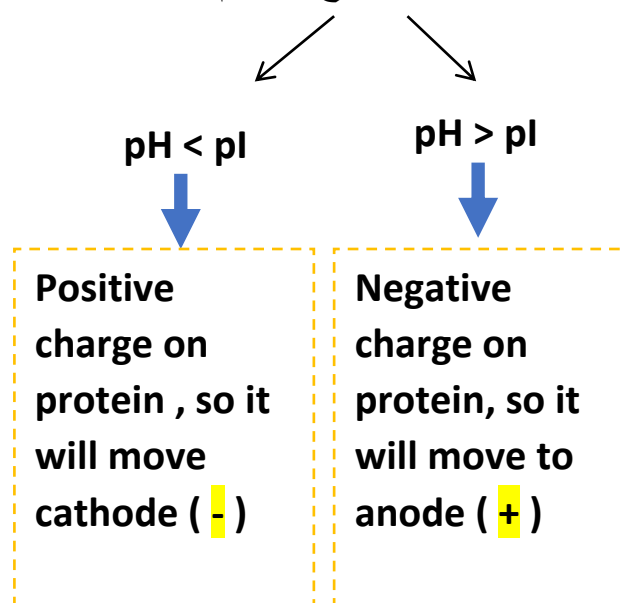


هي عملية انتقال او فصل للبروتين عن طريق التيار الكهربائي (عن طريق شحن البروتين)



According to **Buffer** (منظم للPH)

يعني لما نخط البروتين جوا الخلية الكهربائية رح نخطهم بمحلول اسمو Buffer



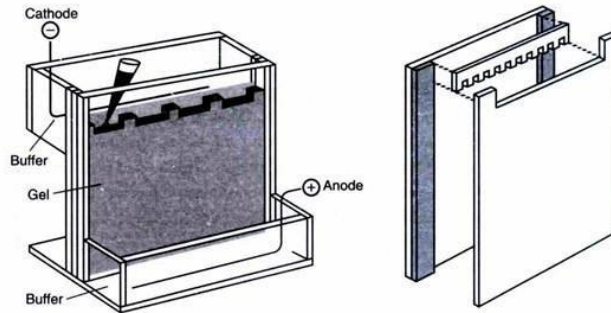


Fig. 8.1: A typical vertical gel apparatus

Stationery phase: (عملية تثبيت البروتين جوا ال Buffer)

The most 4 important examples about Stationery phase :

- 1: sodium dodecyl sulfate (SDS)
- 2: polyacrylamide gel
- 3: agrose
- 4: cellulose acetate

Steps ➡ Application of sample (تجهيز العينة)

➡ Passing the electric current (توصيل التيار الكهربائي)

➡ Fixation of proteins by an acid and alcohol

➡ Staining of proteins by comassie brilliant blue

نصبغه لان البروتينات
Colorless

➡ Destaining by mixture of CH_3COOH , H_2O and CH_3OH

➡ Choice of buffer depend on the isoelectric points of proteins
($\text{PH} > \text{PI}$ OR $\text{PH} < \text{PI}$)

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QUESTIONS

1) Which of the following statement is true about electrophoresis ?

- A) the separation of amino acids in it depend on the charge
- B) the electric field will be switched on after fixation of the amino acid sample
- C) the ninhydrine will stain the gel and the samples so destaining must be done
- D) at a PH value higher than the isoelectric point the sample will migrate toward the positive side

2) electrophoresis does not depend on the electric field ?

- A) True
- B) false

3) if the molar mass of a protein was too small so the velocity of the protein will be ?

- A) too slow
- B) too fast

4) if the protein moves to the anode that means :

- A) $pH > pI$ and the charge is +
- B) $pH < pI$ and the charge is -
- C) $pH > pI$ and the charge is -
- D) $pH < pI$ and the charge is +

5) if the $pH < pI$, which of the following will be true ?

- A) the charge will be +
- B) the charge will be -
- C) the protein will move toward cathode
- D) the protein will move toward anode
- E) A+C
- f) B+D

6) which of the following is considered as an example of stationary phase ?

- A) sodium dodecyl sulfate
- B) poly acrylamide gel
- C) agarose
- D) cellulose acetate
- E) all of the above

7) which of the following we use it when we do fixation for proteins ?

- A) acid
- B) base
- C) alcohol
- D) A+B
- E) A+C

8) we stain proteins by :

- A) Coomassie brilliant blue
- B) CH_3COOH
- C) H_2O
- D) CH_3OH

9) we destain proteins by :

- A) CH_3COOH
- B) H_2O
- C) CH_3OH
- D) All of the above



1-D	2-B	3-B	4-C	5-E	6-E	7-E	8-A	9-D
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